

Exp 4: Data Modelling and Form Design

1. Aim

To design a structured relational data model, construct an Entity-Relationship (ER) diagram, and implement a custom low-code **Service Request Form** in Zoho Creator utilizing varied field types, mandatory constraints, personal data (PII) privacy settings, geographical location capture, and Deluge form scripting definitions.

2. Procedure

2.1 Data Model Specification

The `Service_Request_Form` is designed to capture vehicle maintenance and repair requests. The data model incorporates 13 fields representing varied data primitives (Text, Name, Email, Address with GPS, Phone, Number, Date, Time, Picklist, Radio Buttons, Multi-select List, Checkboxes).

#	Field Name	Field Type	Constraints / Rules	PII Flag	Description & Allowed Values
1	Owner_Name	Name	Mandatory (must have)	Personal Data	First Name and Last Name composite field. Prefix/Suffix hidden.
2	Email_Address	Email	Max 80 chars	Personal Data	Owner contact email with built-in format validation.
3	Pickup_Address	Address	GPS Coords Captured	Personal Data	Sub-fields: Line 1, Line 2, City, State, Postal Code, Country, Lat/Long.
4	Contact_Number	Phone Number	Phone Format	Personal Data	Owner primary telephone/mobile number for notifications.
5	Vehicle_Registration_Number	Single Line Text	Alphanumeric	No	Unique vehicle license plate identifier (e.g., TN-01-AB-1234).
6	Issue_Description	Multi Line Text	Height: 100px	No	Text area for owner to describe vehicle symptoms or problems.
7	Odometer_Reading	Number	Integer (≥ 0)	No	Current vehicle distance reading in kilometers.
8	Preferred_Service_Date	Date	Allowed: 0–6	No	Scheduled date for vehicle pickup or service drop-off.
9	Preferred_Dropoff_Time	Time	hh:mm:ss	No	Specific drop-off time slot requested by the customer.
10	Vehicle_Type	Picklist	Dropdown	No	Values: {"Hatchback", "Sedan", "Other"}.
11	Service_Type	Radio Buttons	Single Select	No	Values: {"Routine Service", "Breakdown Repair", "Accident Repair"}.
12	Additional_Services_Requested	Multi-select List	Height: 60px	No	Values: {"Car Wash", "Interior Cleaning", "Tyre Rotation", "Wheel Alignment", "AC Service"}.

#	Field Name	Field Type	Constraints / Rules	PII Flag	Description & Allowed Values
13	Authorize_Estimate_And_Repair	Checkboxes	Mandatory (must have)	No	Legal authorization choice: {"I authorize the service center to prepare an estimate..."}.

2.2 Entity-Relationship (ER) Diagram

Figure 1 illustrates the conceptual Entity-Relationship structure of the SERVICE_REQUEST_FORM entity, depicting its composite attributes, PII flags, mandatory constraints, and relationships with surrounding automotive service ecosystem entities.

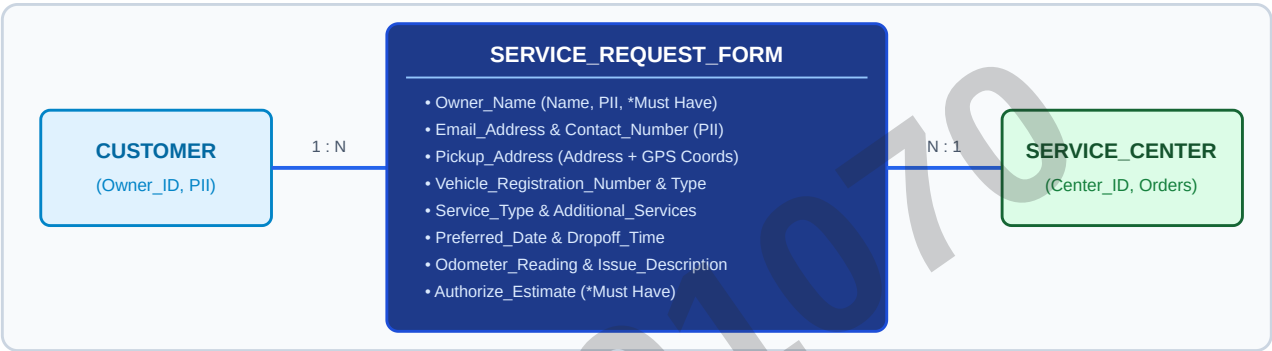


Figure 1: Entity-Relationship (ER) Diagram for Service Request Form Entity

2.3 Zoho Creator Form Building Steps

1. Log into **Zoho Creator Console** and navigate to application workspace *Vehicle Service Management*.
2. Click on **DEVELOP** tab and select **Forms** → **+ Create New Form**.
3. Name the form `Service_Request_Form` and set success message to "Data Added Successfully!".
4. From left-hand **Fields Palette**, drag and drop required field components onto the form canvas.
5. Configure individual field properties in the right-hand Inspector Panel:
 - Set `Owner_Name` property `must have = true` (Mandatory) and enable `Personal Data = true`.
 - Enable `Personal Data = true` on `Email_Address`, `Pickup_Address`, and `Contact_Number` for PII compliance.
 - Enable `capture_coordinates = true` on `Pickup_Address` to capture latitude and longitude.
 - Configure dropdown values for `Vehicle_Type` (Hatchback, Sedan, Other).
 - Configure radio button options for `Service_Type` (Routine Service, Breakdown Repair, Accident Repair).
 - Configure multi-select list choices for `Additional_Services_Requested` (Car Wash, Interior Cleaning, Tyre Rotation, Wheel Alignment, AC Service).
 - Set mandatory checkbox rule on `Authorize_Estimate_And_Repair`.
6. Configure form action buttons: Submit on add, and Update / Cancel on edit.

2.4 Deluge Form Script Definition

The underlying Deluge Script representation for Exp4_Service_Request_Form.ds is provided below:

```
form Service_Request_Form
{
    success message = "Data Added Successfully!"

    Section ( type = section, row = 1, column = 0, width = medium )

    must have Owner_Name
    (
        type = name, displayname = "Owner Name",
        first_name ( type = first_name, displayname = "First Name" ),
        last_name ( type = last_name, displayname = "Last Name" ),
        row = 1, column = 1, width = medium, personal data = true
    )

    Email_Address ( type = email, displayname = "Email Address", maxchar = 80, personal data = true )
    Pickup_Address ( type = address, displayname = "Pickup Address", capture_coordinates = true,
personal data = true )
    Contact_Number ( type = phonenumber, displayname = "Contact Number", personal data = true )
    Vehicle_Registration_Number ( type = text, displayname = "Vehicle Registration Number" )
    Issue_Description ( type = textarea, displayname = "Describe the Issue", height = 100px )
    Odometer_Reading ( type = number, displayname = "Odometer Reading (km)" )
    Preferred_Service_Date ( type = date, displayname = "Preferred Service Date", allowedays =
0,1,2,3,4,5,6 )
    Preferred_Dropoff_Time ( type = time, displayname = "Preferred Drop-off Time", timedisplayoptions =
"hh:mm:ss" )
    Vehicle_Type ( type = picklist, displayname = "Vehicle Type", values =
{"Hatchback","Sedan","Other"} )
    Service_Type ( type = radiobuttons, displayname = "Service Type", values = {"Routine
Service","Breakdown Repair","Accident Repair"} )
    Additional_Services_Requested ( type = list, displayname = "Additional Services Requested", values
= {"Car Wash","Interior Cleaning","Tyre Rotation","Wheel Alignment","AC Service"} )

    must have Authorize_Estimate_And_Repair
    (
        type = checkboxes, displayname = "Authorization",
        values = {"I authorize the service center to prepare an estimate and proceed with the
service/repair"}
    )

    actions {
        on add { submit ( type = submit, displayname = "Submit" ) }
        on edit { update ( type = update, displayname = "Update" ), cancel ( type = cancel, displayname
= "Cancel" ) }
    }
}
```

2.5 Screenshot: Form Creation Workspace



[Screenshot Placeholder: Form Builder Workspace]

Zoho Creator DEVELOP console showing Service_Request_Form canvas, field palette, and field properties panel.

Figure 2: Zoho Creator Form Builder interface with configured fields and mandatory PII flags

3. Output

The constructed Service_Request_Form was deployed and tested within the live application interface. Upon accessing the form, entering valid record details, and clicking the **Submit** button, Zoho Creator validates mandatory field rules (Owner Name and Authorization), enforces PII encryption, captures geographical GPS location coordinates for the pickup address, and displays the configured success message banner "Data Added Successfully!".



[Screenshot Placeholder: Form Submission Output]

Live working view of Service Request Form displaying populated fields and "Data Added Successfully!" notification banner.

Figure 3: Live Service Request Form output demonstrating data entry validation and successful record creation

4. Result

The data model and Entity-Relationship (ER) diagram for the **Service Request Form** were successfully created, and the interactive form was fully built, configured, and validated within Zoho Creator in accordance with the specified field types, mandatory rules, and personal data privacy requirements.